

PYTHON PROJECT REPORT

File Organizer using Python

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Internship	Free Summer Internship in Python – 2026
Project Title	File Organizer
Technology	Python
Duration	4 Weeks
Repository	upskillcampus

1. Project Overview

The File Organizer is a Python-based utility that organizes files in a selected folder automatically. It checks file extensions and moves files into suitable folders such as Images, Documents, Videos, Audio, Archives, and Others. The project demonstrates practical use of Python file handling, conditions, loops, functions, and the os and shutil modules.

2. Objectives

- Understand basic Python programming and program structure.
- Practice working with folders, files, paths, and extensions.
- Use conditions and loops to classify files.
- Automate a practical file-management task.
- Learn basic testing and GitHub submission workflow.

3. Tools and Technologies

Language: Python

Modules: os, shutil

Environment: Python / Command Prompt

Version Control: GitHub

Week 1 – Introduction and Project Planning

Activities and Progress: Reviewed internship requirements and the weekly-report format. Studied introductory Python concepts including variables, input/output, conditions, loops, and functions. Explored beginner-friendly project ideas and selected File Organizer. Planned the file categories and overall program flow.

Challenges and Hurdles: The main challenge was understanding how Python can interact with files and folders. The project was divided into small steps and basic examples were reviewed.

Lessons Learned: Built a basic understanding of Python syntax, project planning, and practical automation.

Week 2 – Python File Handling

Activities and Progress: Studied the os and shutil modules for file and folder operations. Practiced checking paths, identifying file extensions, creating folders, and moving files. Developed the initial classification logic for different file types.

Challenges and Hurdles: File paths and the difference between files and folders required attention. A separate sample folder was used for safer testing.

Lessons Learned: Learned practical file handling, path checking, extension matching, and safe testing.

Week 3 – Project Development and Testing

Activities and Progress: Developed the File Organizer using functions, loops, conditions, os, and shutil. Added categories for Images, Documents, Videos, Audio, Archives, and Others. Added automatic creation of category folders and tested the program with sample files.

Challenges and Hurdles: The main challenge was ensuring correct classification and folder creation before moving files. Different extensions were tested and the logic was adjusted.

Lessons Learned: Improved problem-solving, debugging, and systematic testing skills.

Week 4 – Finalization and Submission Preparation

Activities and Progress: Reviewed the complete program and project structure. Checked the project description, objectives, working process, and expected output. Prepared the report and organized files for submission. Reviewed the GitHub repository, code-file, PDF-report, and link requirements.

Challenges and Hurdles: The final submission process required careful attention to repository naming, file naming, PDF format, and report links.

Lessons Learned: Learned the importance of documentation, version control, organized submission, and technical presentation.

4. Project Working Process

1. User enters the path of the folder to organize.
2. Program checks whether the folder is valid.
3. Required category folders are created automatically.
4. Each file extension is checked.
5. Files are moved to the matching category folder.
6. Unsupported file types are moved to Others.
7. A completion message is displayed.

5. Project Outcome

The project provides a simple way to organize files automatically based on their types. It demonstrates how Python can solve a practical day-to-day problem through automation and provides experience in programming, testing, debugging, and documentation.

6. Future Enhancements

- Add a graphical user interface for selecting folders.
- Allow users to customize categories and extensions.
- Add a log of moved files.
- Add duplicate-file detection and confirmation before moving files.

7. Conclusion

The four-week project provided a practical introduction to Python programming and automation. The work progressed from basic concepts to file handling, project development, testing, documentation, and submission preparation. The File Organizer demonstrates how programming can be applied to a useful real-world problem.

Before submission: Replace the Student Name field and add your actual GitHub code and report links.